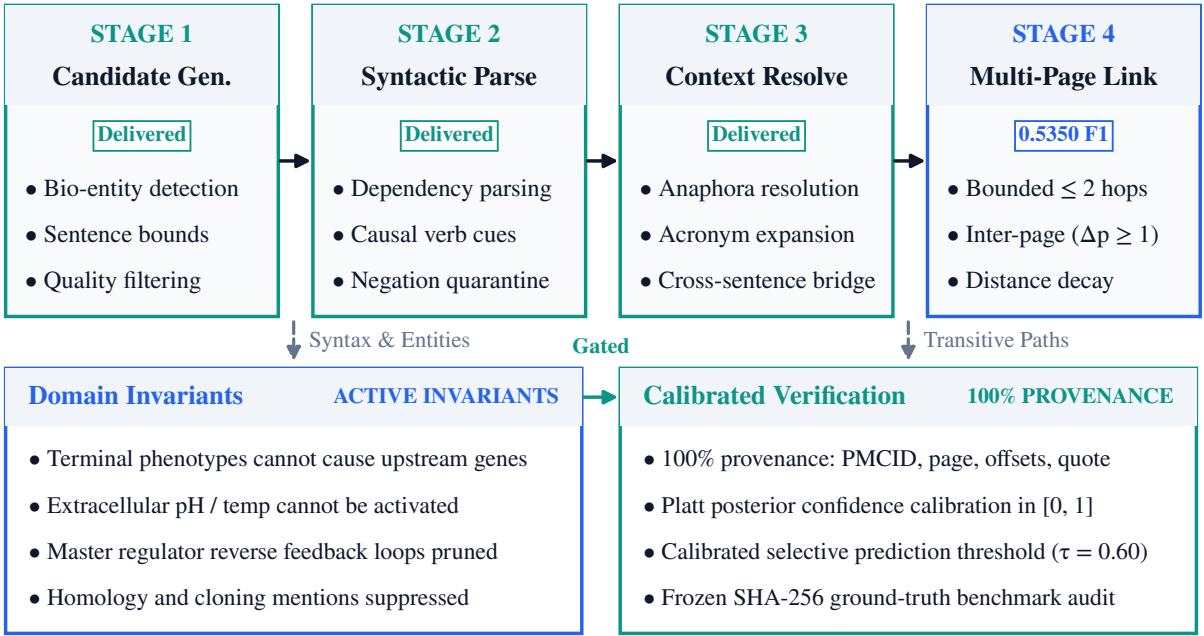
STAGE 1
Candidate Gen.
Delivered
• Bio-entity detection
• Sentence bounds
• Quality filtering
STAGE 2
Syntactic Parse
Delivered
• Dependency parsing
• Causal verb cues
• Negation quarantine
STAGE 3
Context Resolve
Delivered
• Anaphora resolution
• Acronym expansion
• Cross-sentence bridge
STAGE 4
Multi-Page Link
0.5350 F1
• Bounded ≤ 2 hops
• Inter-page (Δp ≥ 1)
• Distance decay
Syntax & Entities
Gated
Transitive Paths
Domain Invariants
ACTIVE INVARIANTS
• Terminal phenotypes cannot cause upstream genes
• Extracellular pH / temp cannot be activated
• Master regulator reverse feedback loops pruned
• Homology and cloning mentions suppressed
Calibrated Verification
100% PROVENANCE
• 100% provenance: PMCID, page, offsets, quote
• Platt posterior confidence calibration in [0, 1]
• Calibrated selective prediction threshold (τ = 0.60)
• Frozen SHA-256 ground-truth benchmark audit